

# Aidi Zhang

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## Work

- 04/2026 – Now     **Postdoctoral Scholar**, Stanford University  
Studying the physical mechanism of the atmospheric rivers.
- 09/2024 – 04/2026     **Postdoctoral Scholar**, University of Chicago  
Studying the physical mechanism of the atmospheric rivers.

## Education

- 2017 – 2024     **Ph.D, Mechanical Engineering, University of California, Berkeley, USA**  
Thesis title: *The Fluid Dynamics of Three-dimensional Jovian Vortices including the Great Red Spot*
- 2013 – 2017     **B.Eng., Theoretical and Applied Mechanics, Sun Yat-Sen University, China**  
Thesis title: *The study of two-dimensional Rayleigh-Bénard Convection with different Prandtl number*

## Research Publications

### Journal Articles

- 1 **A. Zhang**, D. Yang, H. Ong, and Z. Tan, “Understanding the evolution of global atmospheric rivers with a vapor kinetic energy framework,” *Geophysical Research Letters*, vol. 53, no. 7, e2025GL119841, 2026.
- 2 **A. Zhang** and P. Marcus, “Stable three-dimensional vortex families consistent with jovian observations including the great red spot,” *Journal of Fluid Mechanics*, vol. 984, A61, 2024.
- 3 **A. Zhang**, P. Marcus, and S. Kim, “A three-dimensional spectral anelastic code for non-uniform shearing, stratified flow,” in preparation for *Journal of Computational Physics*.
- 4 **A. Zhang** et al., “A three-dimensional stable vortex consistent with multiple proxies of jupiter’s great red spot,” in preparation for *Science*.

### Conference Proceedings

- 1 **A. Zhang**, D. Yang, and H. Ong, “How will atmospheric rivers evolve in a changing climate?” In *AGU Fall Meeting Abstracts*, vol. 2024, 2024, A41T–1868.
- 2 **A. Zhang**, P. S. Marcus, I. De Pater, A. Ermakov, and C. Moeckel, “Three-dimensional vortex families consistent with jovian observations including the great red spot,” in *AGU Fall Meeting Abstracts*, vol. 2023, 2023, P23C–3073.
- 3 **A. Zhang** and P. Marcus, “Numerical study of stable planetary three-dimensional vortices with a hollow vorticity core,” in *APS Division of Fluid Dynamics Meeting Abstracts*, 2023, pp. X13–007.

- 4 **A. Zhang** and P. Marcus, "Stable 3-dimensional vortex families consistent with jovian observations including the great red spot," in *AAS Division of Planetary Sciences Meeting joint with EPSC Abstracts*, 2023.
- 5 **A. Zhang**, A. Ermakov, and P. S. Marcus, "Gravity signatures of stable, equilibrial 3d great red spot solutions consistent with observed cloud-level velocities," in *AGU Fall Meeting Abstracts*, vol. 2022, 2022, P32C-1848.
- 6 **A. Zhang** and P. Marcus, "Hydrodynamic stability constraints on the three-dimensional structure of planetary vortices," APS, 2022.
- 7 P. Marcus and **A. Zhang**, "Vertical aspect ratios and longevities of complex vortices and the application to gfd flows and astrophysical vortices," in *APS Division of Fluid Dynamics Meeting Abstracts*, 2021, H24-009.
- 8 **A. Zhang** and P. Marcus, "Longevity of stratified anticyclones with thermal dissipation and cyclones with viscous dissipation and their relevance to jupiter," in *APS Division of Fluid Dynamics Meeting Abstracts*, 2021, T11-011.
- 9 P. S. Marcus et al., "On the shedding of jupiter's red flakes," in *AGU fall meeting abstracts*, vol. 2019, 2019, P13B-3505.
- 10 **A. Zhang** and P. Marcus, "How the great red spot of jupiter stays alive while losing energy through viscous and radiative dissipation," in *APS Division of Fluid Dynamics Meeting Abstracts*, 2019, B13-004.

## Miscellaneous Experience

### Research Experiences

|                   |                                                                                                                                  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 03/2026 - Now     | ■ Research on the fluid dynamics of atmosphere rivers, Stanford University of                                                    |
| 09/2024 - 02/2026 | ■ Research on the fluid dynamics of atmosphere rivers, University of Chicago                                                     |
| 2018 - 2024       | ■ Research on the longevity and three-dimensional structure of the Great Red Spot on Jupiter, University of California, Berkeley |
| 2016 - 2017       | ■ Research on two-dimensional Rayleigh-Bénard Convection with different Pr number, Sun Yat-Sen University                        |

### Awards

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|------|----------------------------------------------------------------------------------------------------|
| 2023 | ■ <b>Chang-Lin Tien Graduate Fellowship</b> , University of California, Berkeley                   |
| 2022 | ■ <b>Robert P. Lin Graduate Fellowship</b> , Space Science Lab, University of California, Berkeley |
| 2020 | ■ <b>Graduate Division Summer Grant</b> , University of California, Berkeley                       |
| 2019 | ■ <b>Graduate Division Summer Grant</b> , University of California, Berkeley                       |
| 2018 | ■ <b>Graduate Division Summer Grant</b> , University of California, Berkeley                       |
| 2015 | ■ <b>Outstanding undergraduate student</b> , Sun Yat-Sen University                                |
| 2014 | ■ <b>Outstanding undergraduate student</b> , Sun Yat-Sen University                                |

## Miscellaneous Experience (continued)

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### Invited talks

- Jun 5<sup>th</sup>, 2024      **Caltech GPS Yuk Lunch Seminar**, Three-Dimensional Vortex Families Consistent with Jovian Observations Including the Great Red Spot (virtual)
- Jun 14<sup>th</sup>, 2025      **“Climate Chedann” Weekly Climate Dynamics Seminar**, Understanding Atmospheric Rivers Using a Global Vapor Energy Budget Analysis (virtual)

### Internships

- 2022      **Robert P. Lin Fellow**, the Space Science Laboratory at University of California, Berkeley